Zigbee Communication with Arduino Practice

Arduino Code: XBee Basic Communication

Transmitter (Sender)

```
#include <SoftwareSerial.h>

// XBee connected on pins 2 (RX) and 3 (TX)
SoftwareSerial xbeeSerial(2, 3);

int temperature = 0;
int sensorPin = A0;

void setup() {
  Serial.begin(9600);     // Debug via USB
  xbeeSerial.begin(9600); // XBee default baud rate
```

```
void loop() {
  // Read temperature sensor (LM35: 10mV per °C)
  int raw = analogRead(sensorPin);
  float voltage = raw * (3.3 / 1023.0);
  temperature = (int)(voltage * 100.0);

  // Build simple message
  String message = "TEMP:" + String(temperature) + "\n";

  // Send via XBee
  xbeeSerial.print(message);

  // Debug output
  Serial.print("Sent: ");
  Serial.print(message);

  delay(5000);  // Send every 5 seconds
}
```

Arduino Code: XBee Receiver

Receiver (Gateway)

```
#include <SoftwareSerial.h>

SoftwareSerial xbeeSerial(2, 3);

void setup() {
  Serial.begin(9600);
  xbeeSerial.begin(9600);
  Serial.println("XBee Receiver Ready");
}

void loop() {
  // Check if data arrived from XBee
  if (xbeeSerial.available()) {
    String received = "";

    // Read complete line
    while (xbeeSerial.available()) {
      char c = xbeeSerial.read();
      received += c;
      delay(2);  // Small delay for buffer
    }
```

```
void processMessage(String msg) {
  msg.trim();
  Serial.print("Received: ");
  Serial.println(msg);

  // Parse "TEMP:25" format
  if (msg.startsWith("TEMP:")) {
    int temp = msg.substring(5).toInt();
    Serial.print("Temperature: ");
    Serial.print(temp);
    Serial.println(" °C");

    if (temp > 30) {
      Serial.println("⚠️ WARNING: High temperature!");
    }
  }
}
```

Arduino Code: XBee API Mode

Why API Mode? More control, structured packets

```
#include <XBee.h>  // Install "XBee" library by Andrew Rapp

XBee xbee = XBee();
XBeeResponse response = XBeeResponse();
ZBRxResponse rx = ZBRxResponse();

void setup() {
  Serial.begin(9600);   // XBee connected to hardware serial
  xbee.setSerial(Serial);
}

void sendData(uint8_t* data, int length) {
  // Send to coordinator (address 0x0000)
  XBeeAddress64 addr64 = XBeeAddress64(0x00000000, 0x0000FFFF);
  uint16_t addr16 = 0xFFFE;
```

```
void loop() {
  // Read response
  xbee.readPacket(500);

  if (xbee.getResponse().isAvailable()) {
    if (xbee.getResponse().getApiId() == ZB_RX_RESPONSE) {
      xbee.getResponse().getZBRxResponse(rx);

      Serial.print("Data from: ");
      Serial.println(rx.getRemoteAddress64().getMsb(), HEX);

      // Print received bytes
      for (int i = 0; i < rx.getDataLength(); i++) {
        Serial.print((char)rx.getData(i));
      }
      Serial.println();
    }
  }
}
```

Zigbee2MQTT: Practical Integration

Connecting Zigbee to the Internet (Popular DIY Approach)

```
Architecture:
  Zigbee devices
      ↓ (Zigbee radio)
  USB Zigbee Coordinator
  (CC2531 or Sonoff dongle)
      ↓ (USB)
  Raspberry Pi running Zigbee2MQTT
      ↓ (MQTT protocol)
  MQTT Broker (Mosquitto)
      ↓
  Home Assistant / Node-RED / Dashboard

Example MQTT message published:
  Topic:   zigbee2mqtt/kitchen_temp_sensor
  Payload: {
    "temperature": 22.5,
    "humidity": 45,
    "battery": 87,
    "linkquality": 142
  }
```

Zigbee2MQTT: Python Subscriber

Reading Zigbee Sensor Data via MQTT

```python
import paho.mqtt.client as mqtt
import json

BROKER = "localhost"
TOPIC = "zigbee2mqtt/#"  # Subscribe to all Zigbee devices

def on_connect(client, userdata, flags, rc):
    print(f"Connected to MQTT broker (code: {rc})")
    client.subscribe(TOPIC)

def on_message(client, userdata, msg):
    topic = msg.topic

    try:
        payload = json.loads(msg.payload.decode())
    except json.JSONDecodeError:
        return  # Skip non-JSON messages

    # Extract device name from topic
    device = topic.replace("zigbee2mqtt/", "")

    # Process temperature sensors
    if "temperature" in payload:
        temp = payload["temperature"]
        humidity = payload.get("humidity", "N/A")
        battery = payload.get("battery", "N/A")

        print(f"[{device}]")
        print(f"  Temperature : {temp} °C")
        print(f"  Humidity    : {humidity} %")
        print(f"  Battery     : {battery} %")

    # Process motion sensors
    elif "occupancy" in payload:
        motion = payload["occupancy"]
        print(f"[{device}] Motion: {'DETECTED' if motion else 'clear'}")

    # Process switches/buttons
    elif "action" in payload:
        action = payload["action"]
        print(f"[{device}] Action: {action}")
```

Zigbee2MQTT: Control Devices via Python

Sending Commands to Zigbee Devices

```python
import paho.mqtt.client as mqtt
import json
import time

BROKER = "localhost"

def control_light(client, device_name, state):
    """Turn a Zigbee light ON or OFF"""
    topic = f"zigbee2mqtt/{device_name}/set"
    payload = json.dumps({"state": state})  # "ON" or "OFF"
    client.publish(topic, payload)
    print(f"Sent {state} to {device_name}")

def set_brightness(client, device_name, brightness):
    """Set brightness 0-254"""
    topic = f"zigbee2mqtt/{device_name}/set"
    payload = json.dumps({"brightness": brightness})
    client.publish(topic, payload)
    print(f"Set brightness {brightness} for {device_name}")

def set_color_temp(client, device_name, color_temp):
    """Set color temperature in Kelvin (2700=warm, 6500=cool)"""
    topic = f"zigbee2mqtt/{device_name}/set"
    payload = json.dumps({"color_temp": color_temp})
    client.publish(topic, payload)

# Connect and control
client = mqtt.Client()
client.connect(BROKER, 1883, 60)
client.loop_start()

time.sleep(1)  # Wait for connection

# Example: Morning automation
print("Morning automation!")
control_light(client, "bedroom_light", "ON")
set_brightness(client, "bedroom_light", 80)      # Dim
set_color_temp(client, "bedroom_light", 3000)    # Warm white
```